



Islamic University of Technology
Department of Computer Science and Engineering

Lab 8: Dynamic Programming

CSE 4404: Algorithms Lab
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Task A. Common Ground

Time Limit: 1 second | Memory Limit: 256 MB

You are given two strings s and t consisting of lowercase English letters. Find the length of their longest common subsequence: the longest sequence of characters that appears, in order, in both strings. The characters of the subsequence do not need to be adjacent in either string.

Input Format

The first line contains the string s . The second line contains the string t .

Output Format

Print a single integer: the length of the longest common subsequence of s and t .

Constraints

- $1 \leq |s|, |t| \leq 3000$
- s and t consist only of lowercase English letters.

Examples

Sample Input	Sample Output
abcde ace	3
abcbdad bdcaba	4

Task B. Museum Heist

Time Limit: 2 seconds | Memory Limit: 512 MB

You have broken into a museum holding n relics. You know the weight and worth of each relic. Your getaway bag can carry a total weight of at most x . What is the maximum total worth you can escape with? You may take each relic at most once.

Input Format

The first line contains two integers n and x : the number of relics and the maximum total weight the bag can carry.

The next line contains n integers $w_1, w_2, \dots, w_n$: the weight of each relic.

The last line contains n integers $v_1, v_2, \dots, v_n$: the worth of each relic.

Output Format

Print one integer: the maximum total worth you can take.

Constraints

- $1 \leq n \leq 1000$
- $1 \leq x \leq 10^5$
- $1 \leq w_i, v_i \leq 1000$

Examples

Sample Input	Sample Output
4 10 3 4 5 8 4 5 6 9	11
3 5 1 3 4 2 4 5	7

Task C. Postage Stamps

Time Limit: 1 second | Memory Limit: 256 MB

A post office has cnt_i stamps of denomination c_i remaining, for $i = 1, \dots, n$. A parcel requires exactly x in postage. Determine the minimum number of stamps needed to make exactly x using the available stock, or determine that it is impossible.

Input Format

The first line contains two integers n and x . Each of the next n lines contains two integers c_i and cnt_i : the value of the i -th denomination and how many such stamps remain.

Output Format

Print the minimum number of stamps needed to make exactly x , or -1 if it cannot be done with the available stock.

Constraints

- $1 \leq n \leq 100$
- $1 \leq x \leq 1000$
- $1 \leq c_i \leq 1000$
- $1 \leq cnt_i \leq 100$

Examples

Sample Input	Sample Output
3 17 2 50 5 50 10 50	3
2 7 5 1 3 1	-1

Marks Distribution

Task	Marks
Task A	30%
Task B	35%
Task C	35%

Practice Problems

Problem	Link
Longest Common Subsequence	AtCoder EDU DP F: https://atcoder.jp/contests/dp/tasks/dp_f
Knapsack 1	AtCoder EDU DP E: https://atcoder.jp/contests/dp/tasks/dp_e
Minimizing Coins	CSES: https://cses.fi/problemset/task/1634